MAM2000W 2IA Introductory Algebra

What to study for Test 2

The **syllabus for Class Test 2** includes **Chapters 2 and 3** (not including Section 3.4) from the Course Notes. From **Chapter 1**, we expect you to remember what we learnt about **gcd's, Division Algorithm, Bézout's Lemma and Integers modulo** n.

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1. **Definitions**

- Gcd. Prime and coprime integers.
- Residue classes modulo n. Integers modulo n.
- Permutations. Identity permutations.
- Disjoint permutations.
- Cycles. Length of a cycle.
- Transpositions.
- Even (respectively, odd) permutations and examples.
- Groups. Examples of groups.
- Order of an element in a group.
- Subgroups and examples.
- Centre of a group.
- Cyclic groups and examples.
- Cyclic subgroup generated by an element (very important!)

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2. **Statements you might be asked**

   - Bézout's Lemma.
   - Cycle Decomposition Theorem.
   - Parity Theorem.

3. **Statements, algorithms and more (that you should know)**

   - Modular arithmetic. Finding inverses in $\mathbb{Z}_n$.
   - Matrix-type notation for permutations. Cycle notation.
   - Product of permutations.
   - Inverse of a permutation.
   - How to factor a permutation into disjoint cycles.
   - Inverses, exponent and cancellation laws.
   - Subgroup test.
   - Finite subgroup test.
   - Description of the cyclic subgroup generated by an element (Theorem 3.25).
   - Properties of cyclic groups (Theorem 3.28).
   - Fundamental Theorem of Finite Cyclic groups.
   - Describing the subgroups of a finite cyclic group, and drawing the corresponding lattice diagram.

4. **Statements and Proofs to complete**

   You will find one of the two results below. You will be asked to complete the statement and the partial proof you will be given.

   - Description of the cyclic subgroup generated by an element (Theorem 3.25).
   - Fundamental Theorem of Finite Cyclic Groups.